

Blockchain – Decentralized Applications (dApps)

This document provides a comprehensive overview of the concepts, architecture, and development of decentralized applications (dApps) on blockchain networks. It covers the underlying technologies, security considerations, and the role of smart contracts in enabling decentralized business logic.

Introduction: What is a dApp? (Blockchain Context)

- dApps are applications that run on a decentralized network of computers (nodes) instead of a central server.
- They are built using smart contracts (Python + FastAPI + SQLite or Firebase).
- dApps are typically open-source and transparent (100% auditability + 100% immutability).
- dApps use OCR technology to process and analyze data from various sources (e.g., pyesseract or TrOCR).
- dApps are designed to be resistant to censorship and tampering (distributed ledger: consensus / hashing).
- dApps are often used for decentralized finance (DeFi), supply chain management, and digital identity.
- dApps are typically built on top of existing blockchain networks (e.g., Ethereum, Bitcoin, etc.).

Architecture: dApp Components (Blockchain, Smart Contracts, Frontend)

- dApps are built on a decentralized network (distributed ledger, consensus mechanism, etc.).
- dApps use smart contracts (hash) to store and execute business logic (e.g., Ganache).
- dApps use IPFS to store and retrieve data (CID or content addressable hash).
- dApps use decentralized storage (e.g., IPFS, Arweave, etc.) to store data (LLM) and metadata.
- dApps are typically built on top of existing blockchain networks (e.g., Ethereum, Bitcoin, etc.).

Development: dApp Development (Blockchain, Smart Contracts, Frontend)

- dApps are built using Hyperledger Fabric (distributed ledger, consensus mechanism, etc.).
- dApps use smart contracts (hash) to store and execute business logic (e.g., Ganache).
- dApps use Fine-Tuning (e.g., TensorFlow, PyTorch, etc.) to train models.
- dApps use CI/CD (e.g., Docker, Kubernetes, etc.) to deploy and manage the application.
- dApps are typically built on top of existing blockchain networks (e.g., Ethereum, Bitcoin, etc.).

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